

Pandas & Data Cleaning

Dataset Used Throughout

Titanic Dataset (titanic.csv) — Kaggle Columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

■ Loading the Dataset

- Q1.** Load titanic.csv into a DataFrame using `pd.read_csv()` and display the first 10 rows.
- Q2.** Display the shape, column names, and data types of the Titanic dataset.
- Q3.** The Titanic CSV file uses comma as delimiter. Reload it explicitly passing the separator and confirm it loads correctly.

■ Early Data Inspection

- Q4.** Use `df.info()` to display column names, non-null counts, and data types of the Titanic dataset.
- Q5.** Use `df.describe(include='all')` and identify which column has the highest standard deviation.
- Q6.** From `df.describe(include='all')`, identify which columns are numeric and which are categorical.
- Q7.** Using `df.info()`, identify how many columns have at least one missing value.

■ Identifying Missing Data

- Q20.** Display the total number of missing values in each column of the Titanic dataset.
 - Q21.** Display the percentage of missing values per column, sorted in descending order.
 - Q22.** Using `notnull()`, filter and display only rows where both 'Age' and 'Cabin' are present.
 - Q23.** Display the total count of missing values in the entire dataset as a single number.
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■ `dropna()`

- Q24.** Drop all rows that have any missing value and print how many rows were removed.
 - Q25.** Drop only the rows where 'Age' is missing using the `subset` parameter.
 - Q26.** Drop only columns that have more than 70% missing values.
 - Q27.** Use the `thresh` parameter to keep only rows that have at least 10 non-null values.
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■ `fillna()`

- Q28.** Fill missing values in the 'Age' column with the median age.
 - Q29.** Fill missing values in the 'Embarked' column with the mode.
 - Q30.** Fill missing values in 'Age' using forward fill and check if any nulls remain.
 - Q31.** Fill missing values in 'Fare' using backward fill.
 - Q32.** Fill missing 'Age' values with the mean age grouped by 'Pclass'.
 - Q33.** Fill missing 'Age' values with the mean age grouped by both 'Pclass' and 'Sex'.
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■ Missingness Mechanisms (Conceptual + Applied)

- Q34.** The 'Cabin' column has ~77% missing values. Analyze whether this is MCAR, MAR, or MNAR. Justify with data evidence.
- Q35.** Check if 'Age' missingness is related to 'Pclass'. Does this suggest MCAR, MAR, or MNAR?

■ Outlier Detection — Boxplot Method

- Q36.** Using IQR boundaries, identify how many outliers exist in the 'Fare' column based on a boxplot analysis.

Q37. Apply the boxplot method on the 'Age' column and list the outlier values found.

■ Outlier Detection — Z-Score Method

Q38. Using the Z-score method (threshold = 3), find all rows where 'Fare' is an outlier.

Q39. Apply Z-score outlier detection on the 'Age' column and display the count of detected outliers.

Q40. Apply Z-score on all numeric columns and report which column has the highest number of outliers.

■ Outlier Detection — IQR Method

Q41. Using the IQR method, detect outliers in the 'Fare' column. Display the lower and upper bounds used.

Q42. Apply IQR-based outlier detection on the 'Age' column and display the percentage of rows that are outliers.

Q43. Apply IQR outlier detection on both 'Age' and 'Fare'. Which column has more outliers?

■ Outlier Treatment — Capping

Q44. Cap outliers in the 'Fare' column at the 5th and 95th percentiles. Verify using describe().

Q45. Apply Winsorization (1st and 99th percentile) on the 'Age' column and compare min/max before and after.

Q46. Cap 'Fare' outliers using IQR method (lower = $Q1 - 1.5 \times IQR$, upper = $Q3 + 1.5 \times IQR$) and display the result.
